

Contribution Journal

Ebuka

My contribution focused on completing the exploratory data analysis notebook, reviewing its outputs, creating the required experiment visualizations, and preparing my written explanations. I also helped other group members understand the assignment and what needed to be included in the submission. My written contribution included answering the knowledge-check questions and reviewing the reflective journal. I explained the purpose of EDA, the passenger characteristics associated with higher survival, and why visualizations provide information beyond summary statistics. Through this collaboration, I practiced interpreting charts, checking results, and explaining the purpose of each stage in the EDA workflow.

Sam

Deconstructing Imputation Strategies: Explored the structural differences between dropping rows, mean imputation, and median imputation. Established a clear framework for selecting the median when dealing with skewed real-world metrics like age, ensuring robustness against extreme outliers.

Clarifying Categorical Transformations: Analyzed the risks of naive integer mapping for categorical variables and reinforced why one-hot encoding is the industry standard. Documented how expanding categories into binary sparse matrices prevents algorithms from inferring false mathematical rankings.

Understanding Optimization Dynamics: Examined the mathematical impact of unscaled features on gradient descent and distance metrics. Mapped out how StandardScaler normalizes variance across columns, preventing high-magnitude features from dominating model training.

Sadia Ravate

I also helped work through the lab when some of the code cells were showing syntax errors because the code was all placed on one line. I went through the affected cells, separated the code into the correct lines, and reran them until the visualizations worked. I also

worked on the EDA questions and interpreted the charts to understand the survival patterns in the Titanic dataset.